

IIT Madras

ONLINE DEGREE

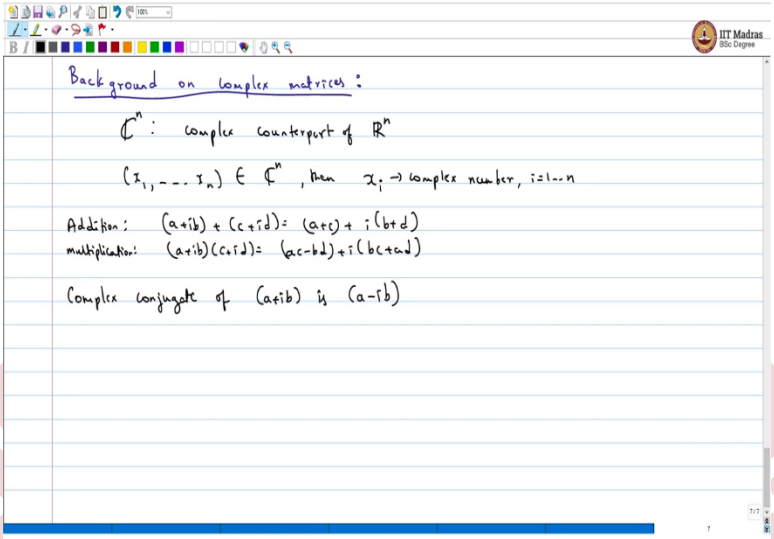
Machine Learning Foundation
Professor Prashanth L. A.
Department of Computer Science and Engineering
Indian Institute of Technology, Madras
Complex Matrices

Okay welcome to week 5 of Machine Learning Foundations. We will be continuing the discussion of linear algebra all through this week, and one more week after this. The goal of this week, the lectures in this week is to finally get to the most important theorem in linear algebra which is the spectral theorem, and we will get to this in a more general form, not necessarily real vector spaces but we will look at this theorem in the context of a complex vector space.

And the important class of matrices that, that generalize real symmetric matrices is hermitian matrices. So, we will show, we will prove spectral theorem in the context of a hermitian matrix to show that it is the main result, in very simple term says that every hermitian matrix is orthogonally diagonalizable.

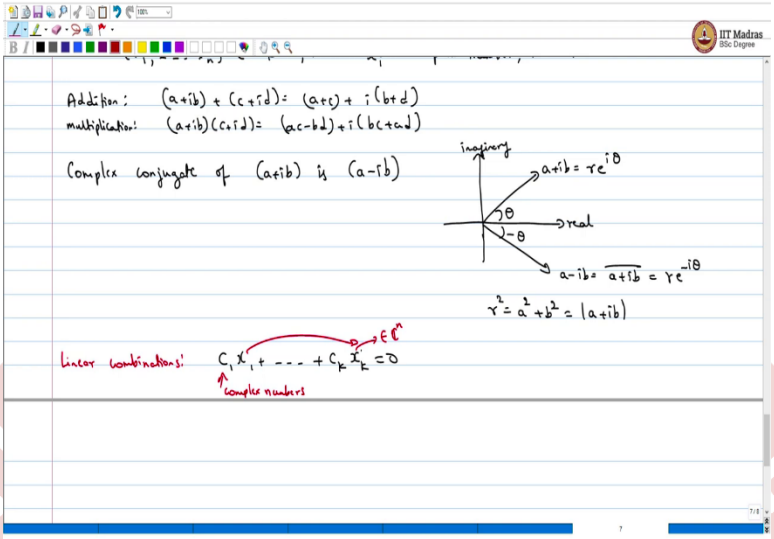
So, as a gentle *t, in this lecture we will look at complex matrices, complex vectors and complex matrices and try to understand how we can define inner product between two complex vectors in the more natural form. And then we will also define what it means to take the conjugate transpose of a complex matrix and use this notion of conjugate transpose to define a hermitian matrix.

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So, we will denote this by Cn. This is the complex counterpart of the usual Euclidean space, n dimensional Euclidean space Rn. So, when I say x 1 up till x n belongs to this, then what I am saying is each x i is a complex number for i = 1 to n. Now, I mean, this is something that you may already know, how do you, addition of complex numbers. If I have a + ib + c + id then I add the real and imaginary parts. And multiplication also, this will be ac - bd + i into bc + ad. This is something that I hope you are all familiar with, so I am not going to spend too much time on this. Now, the important thing to understand is its complex conjugate. Complex conjugate of a complex number a + ib is a - ib.

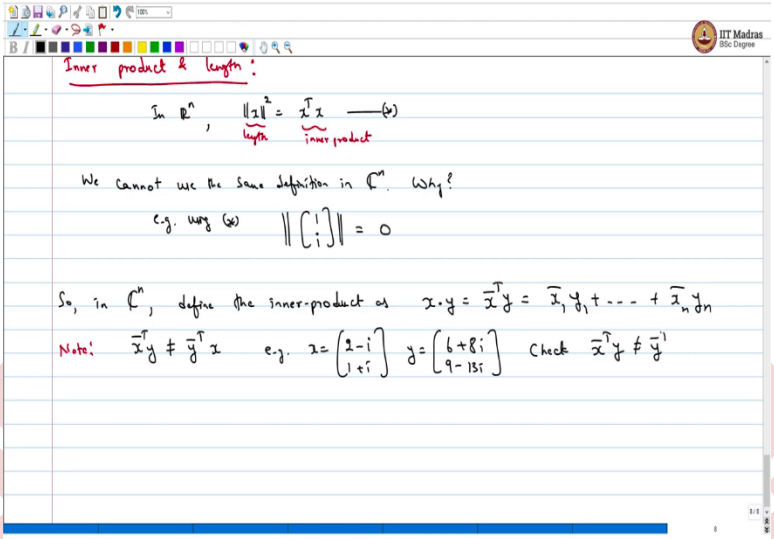
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Way to understand this is, if you have a complex number you can have, use the polar coordinates to represent a + ib as r e power iθ θ and now this is the real axis and this is the imaginary axis. And now the conjugate would be written as a - ib, this is the conjugate, denoted like this, and that can be represented as r e power - i θ, where r square is, this is nothing but the length of this vector.

Now, you can take linear combinations as before, linear combinations with a slight tweak. So, if I write c1 v1 +, let me use x1, cn xn = 0, let us say k, if I take k vectors, xk = 0, then the difference from before is these scalars could be complex numbers. And these belong to Cn.

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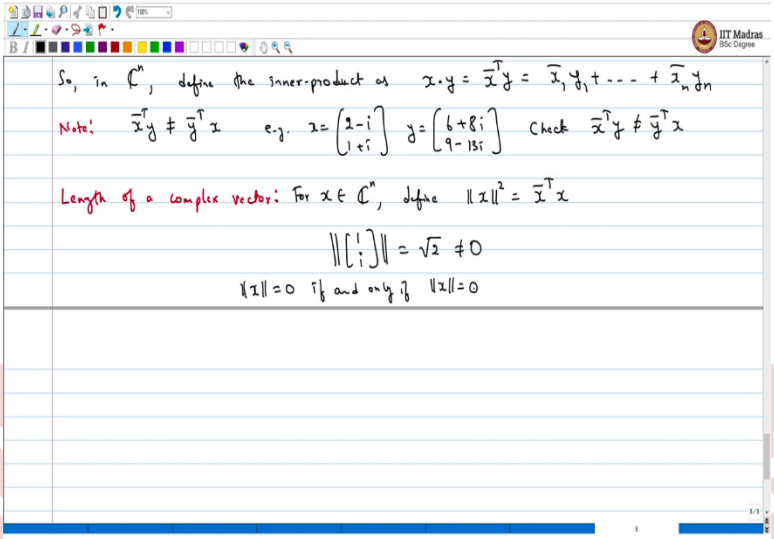


Now, the next bit of background to cover is inner product, and also the length, the concept of length of a vector in the complex case. So, recall in Euclidean space this was your dot product, this was the inner product, and that was used to define the length of a vector. Now, we cannot use the same definition in the complex case. Why? Because it does not make intuitions. Take an example. For example, using * what would be the length of 1i? This would just be 1 square + i square, which will just be 0.

So, in the case of a complex vector space, define the inner product a little differently using what are known as conjugates as, if I want x dot product with y that will be x bar transpose y, which is like I take the conjugate of the first one and multiply with the second one.

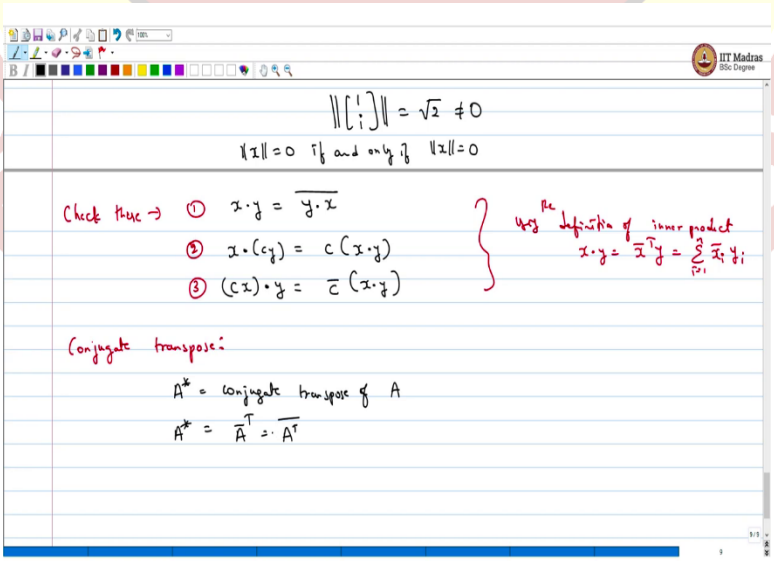
The important thing is this relation, very important note, using this we can easily see that x bar transpose y is definitely not equal to y bar transpose x. I mean if you want, take for example x is 2 - I, 1 + i and y is, I mean if you take real numbers, if you take real x and y then obviously this relation holds, so you need to take complex numbers. Now, check x bar transpose y is not equal to y bar transpose.

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Now, how do I length, define the length of a complex vector? The way I do it is, x belongs to this, define the length as x bar transpose x. Now, with this, the length of 1 , I, what would this be? This woold not be 0 when you take the conjugates, but you can check that this turns out to be square root 2, and its not a 0-length vector, which is, which coincides with our intuition. So, 1 , i is not a 0 length vector and so our definition if that vector is 0, otherwise we can, otherwise we will not get the zero length vector.

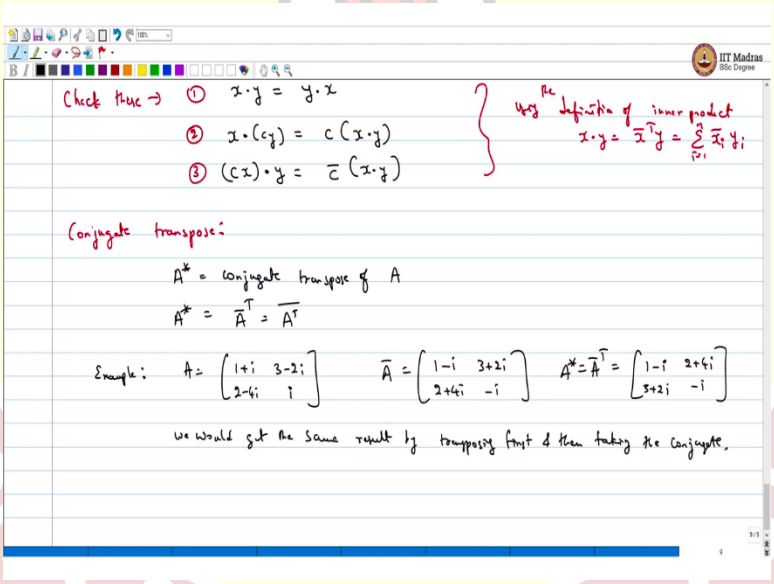
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Now, you can check this, I am going to just check these. 1, this can be done by definition. If I take x dot y is y dot x conjugate and if I do x dot some constant times y, then this is going to be just that constant times x dot y. Simply because you are taking the conjugate of the first component in the inner product and, but the same is not true of this one. When you do it like this the constant comes out as a conjugate. So, check this. This can be verified by definitions. Check this using definition of dot product, inner product, that is, x dot y = x bar transpose y summation x bar i do y = 1 to i.

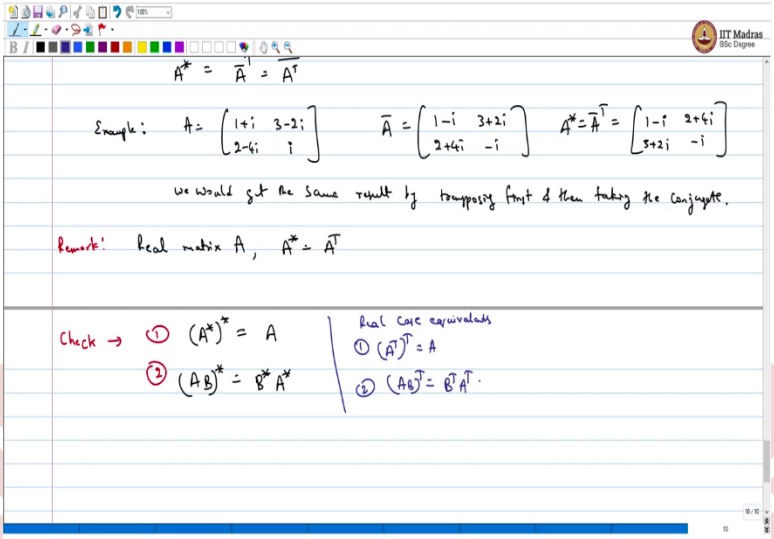
Now, given this we define what is called a conjugate transpose. Everybody knows transpose, so here you add the conjugate. So, so A * is that matrix, which is the conjugate transpose of A, and it is defined as A * =, you can do this in either order, and they lead to the same solution.

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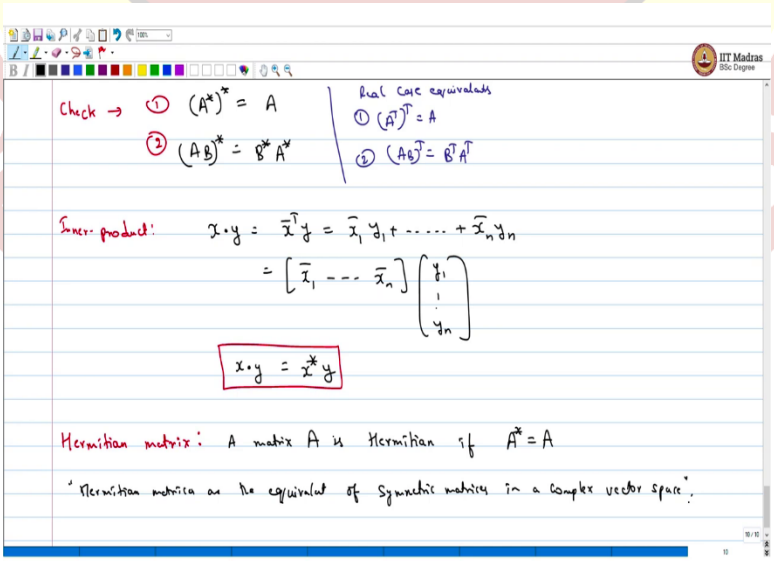
As an example, suppose A is this 1 + I, 3 - 2i, 2 - 4i and i. If you have this then what this A bar, A bar is just 1 - i, 3 + 2i, 2 + 4i, 1 - i. And A bar transpose, A * = A bar transpose =, we would get the same result by transposing first and then taking the conjugate, we will get the same result, that would not change.

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Important remark, for a real matrix, A * is nothing but A transpose because for a real number it is its own conjugate. Now, some claims, again, this can be checked. Check, these are just the equivalent of transpose that we have. If I do A * and do another *, what I recover is this, and AB * is B * A *. Now, the real, in the real case what you would get is, the equivalent is equivalents is A transpose transpose = A and A B transpose is B transpose A transpose.

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Just reiterating the inner product, so to have it defined in one place, x dot y is x bar transpose y. So, this is x1 bar y1 …, xn bar yn. This can be written as x1 bar until xn bar y1 to yn. But this is the same as x * into y. So, x dot y = this. The last bit of definition before we close this lecture is what is called as a hermitian matrix. Matrix A is hermitian, this is the equivalent of symmetric matrices. If A * = A. Hermitian matrices are the equivalent of symmetric matrices in a complex vector space.

For a real symmetric matrix you have A transpose = A, and for hermitian matrix if I take the conjugate transpose, it is, it is the same. So, quick recap before I end this lecture. What did we look at the, the, what are the contents of this lecture? At a very high level we understood how we can define the inner product between two complex vectors and this involved some change. In particular, we had to do a conjugate operation instead of just the transpose. We had the conjugate operation in addition.

And the second thing that we looked at was conjugate transpose, the notion of conjugate transpose, which when applied to matrices, we used, we applied it to a matrix, and if a matrix and its conjugate transpose coincide that is the so-called Hermitian matrix. And it is very, it is a very important subclass of complex matrices which we will be concentrating on, because these are matrices that will be orthogonal diagonalizable, which will become apparent as we go on in this week, in the forthcoming lectures.